

Patient Shortage in Histopathology: Does Coverage-Maximizing Selection Help Five-Patient Cohorts?

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Abstract

TCGA-UT patch classifiers trained on five patients per class are 9.52 percentage points less accurate than classifiers trained on twenty patients with the same patch budget. Coverage, the distance from held-out patients to their nearest training patient, is a candidate explanation of the part of this gap that effective support leaves open, but in random cohorts it changes almost only with the patient count. This experiment varies coverage at a fixed count of five patients with 32 patches each. A cohort chosen by coverage-maximizing selection is compared with the randomly drawn cohort that shares its focal patient. Before any classifier was trained, a manipulation check showed that selection achieves 65 to 68 percent of the coverage gained when fifteen random patients join the five. The selected cohort reaches 67.87 percent patient-macro balanced accuracy, 5.52 points above the random cohort (test-patient 95% interval: 4.83 to 6.23), and recovers 58 percent of the advantage of twenty over five random patients. The gain is positive in all fifteen split-draw combinations and concentrates on test patients whose coverage improves most. Coverage-maximizing selection therefore mitigates patient shortage at five patients, although the selected patients are also less similar to one another than random ones.

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1 Motivation

At 160 patches per TCGA-UT class, twenty patients contributing eight patches each outperform five patients contributing 32 patches each by 9.52 percentage points of patient-macro balanced accuracy (Queisler, 2026c). Effective support, which discounts patches by their correlation within a patient, explains less than half of this gap: each doubling of patients still adds 2.62 points at equal effective support, about 5.2 points over the two doublings (Queisler, 2026d). Doubling tissue source sites at a fixed patient count adds 1.06 points (Queisler, 2026e).

At twenty patients, cohort composition matters. A cohort built from one patient and its nineteen nearest neighbours in the Virchow2 embedding is 8.30 points less accurate than a random cohort of the same size, and a cohort chosen by coverage-maximizing selection is 1.04 points more accurate (95% interval: 0.44 to 1.66) (Queisler, 2026a). A relation fitted to random cohorts of five to twenty patients links accuracy to coverage distance and to similarity-adjusted effective support. With this relation, the residual benefit of additional patients falls from 2.62 to 0.79 points per doubling, but its interval of -1.25 to 2.82 points crosses both practical thresholds (Queisler, 2026b). In random cohorts, coverage distance is nearly fixed by the patient count, so coverage and patient count cannot be separated. The same relation predicts a selection gain of 2.10 points at twenty patients, about twice the observed gain, which fits diminishing returns once twenty random patients already cover most held-out patients.

Five patients leave more coverage to gain. Five random patients cover validation patients at a mean distance of 0.385 to 0.387, twenty random patients at 0.292 to 0.296 (Queisler, 2026a). Five patients are also the allocation in which patient shortage was first observed. This report therefore asks: *with five patients and 32 patches per class, does a cohort chosen by coverage-maximizing selection train a more accurate classifier than a randomly drawn cohort?* The comparison varies coverage at a patient count other than twenty and tests directly whether selection recovers part of the breadth benefit when patients are scarce.

2 Study design

Only the choice of five patients changes. The design inherits from Queisler (2026a) the 30 eligible TCGA-UT cancer types, the frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), the three fixed patient-disjoint splits with their natural validation and test partitions, the nested round-robin patch order, and the readout: multinomial logistic regression with its regularization strength selected on the validation partition for every classifier. It also inherits the patient distance. For split r and class c , $\bar{\mathbf{x}}_{irc}$ is the mean feature vector of patient i ’s patches of the class, $\boldsymbol{\mu}_r$ the average of these means over all classes, and

$$\mathbf{e}_{irc} = \frac{\bar{\mathbf{x}}_{irc} - \boldsymbol{\mu}_r}{\|\bar{\mathbf{x}}_{irc} - \boldsymbol{\mu}_r\|}, \quad d(i, j) = 1 - \mathbf{e}_{irc}^\top \mathbf{e}_{jrc}. \quad (1)$$

Allocation	G	m	Patients
Random	5	32	five anchor patients, including the focal patient
Dispersed	5	32	focal patient, 4 coverage-maximizing patients

Table 1: The two allocations share the focal patient, the patient count G , the patches per patient m , the budget of 160 patches per class, and therefore the effective support. They differ only in how the other four patients are chosen.

Table 1 lists the allocations. Effective support depends only on G , m , and the intraclass correlations, so it is identical by construction.

2.1 Allocation rules

The pool $\mathcal{E}_{rc}$ contains the training patients with at least 32 patches of class c ; pools hold between 30 and 514 patients. For each split, class, and draw $d \in \{1, \dots, 5\}$, the five *anchor patients* are drawn with the five-patient sampling rule of Queisler (2026c), as reused by Queisler (2026a), whose split-0 class order is used for every split. The anchors therefore coincide with the breadth–depth grid’s stored five-patient draw only in split 0; in the other two splits the class order differs, so the random allocation is fitted fresh at $G = 5$, $m = 32$ in all three splits (30 classifiers in total) rather than reused from the grid’s stored classifiers.

The *focal patient* f is the anchor drawn uniformly at random in Queisler (2026a). The dispersed allocation starts from $S_0 = \{f\}$ and adds four patients greedily by *coverage-maximizing selection*, known as facility location in operations research:

$$S_{k+1} = S_k \cup \left\{ \arg \min_{j \in \mathcal{E}_{rc} \setminus S_k} \frac{1}{|\mathcal{E}_{rc}|} \sum_{i \in \mathcal{E}_{rc}} \min_{l \in S_k \cup \{j\}} d(i, l) \right\}, \quad k = 0, \dots, 3. \quad (2)$$

Each step adds the pool patient that most shortens the mean distance from pool patients to their nearest selected patient. Because each step depends only on the patients already selected, the dispersed allocation equals the first five patients of the twenty-patient dispersed allocation of Queisler (2026a). The rule uses only unlabelled training embeddings, not tissue source sites, validation patients, or test patients. Ties are broken by patient identifier, and every patient contributes the first 32 patches of its round-robin order.

2.2 Manipulation check

A prespecified manipulation check tests whether selection moves coverage by a substantial amount before any classifier is trained. The *coverage distance* of a patient set is the mean distance from the class’s validation patients $\mathcal{V}_{rc}$ to their nearest training patient,

$$r_{rc}(\mathcal{P}) = \frac{1}{|\mathcal{V}_{rc}|} \sum_{i \in \mathcal{V}_{rc}} \min_{j \in \mathcal{P}} d(i, j). \quad (3)$$

Let r_5 be the coverage distance of the five anchors, r_{dis} that of the dispersed allocation, and r_{20} that of the twenty-patient random allocation of Queisler (2026a), which adds fifteen random pool patients to the same anchors. Each is averaged over classes and draws within a split. The difference $r_5 - r_{20}$ is the coverage gained when fifteen random patients join the five anchors. The *selection ratio*

$$\rho_{\text{sel}} = \frac{r_5 - r_{\text{dis}}}{r_5 - r_{20}} \quad (4)$$

expresses the coverage change of selection as a fraction of this gain. Fitting proceeds only if $\rho_{\text{sel}} \geq 0.5$ in all three splits; this threshold was fixed before the ratio was computed. The class-level ratios, the mean pairwise distance within each cohort, and the share of non-focal patients from the focal patient’s tissue source site are recorded without a threshold.

The check passes (Table 2). Four selected patients shorten the coverage distance from about 0.386 to 0.324, which is below the 0.333 of ten random patients (Queisler, 2026b). The ratio lies between 0.65 and 0.68, and the class-level ratio reaches 0.5 in 77 of 90 class–split combinations. Selection does not favour the focal patient’s tissue source site: 17 to 18 percent of the four added patients share it, against 15 to 17 percent of the four other anchors.

3 Evaluation

The evaluation reads one primary contrast and describes where and how much of the breadth benefit it recovers.

Split	Coverage distance				Classes with ratio ≥ 0.5
	r_5	r_{dis}	r_{20}	ρ_{sel}	
1	0.387	0.325	0.296	0.68	26 of 30
2	0.385	0.323	0.293	0.68	26 of 30
3	0.385	0.324	0.292	0.65	25 of 30

Table 2: Four coverage-maximizing patients achieve about two thirds of the coverage gained when fifteen random patients join five. Coverage distances are mean cosine distances from validation patients to their nearest training patient, averaged over the 30 classes and five draws. Mean pairwise distances within a cohort are 0.58 to 0.60 for the anchors and 0.60 to 0.63 for dispersed allocations. Selecting all five patients without a focal patient would raise ρ_{sel} to 0.73 to 0.75.

3.1 Selection gain at five patients

Accuracy is patient-macro balanced accuracy over all 30 classes: patch recall is averaged within each test patient and true class, then across patients within each class, and finally across classes. Averaging three splits and five draws gives A_{ran} and A_{dis} in percent. The primary endpoint is the *selection gain*

$$\Delta_{\text{sel}} = A_{\text{dis}} - A_{\text{ran}}, \quad (5)$$

the accuracy won when coverage-maximizing selection replaces random sampling of five patients. It holds patients, patches per patient, budget, and effective support fixed.

Four secondary analyses carry no decision threshold. First, the *recovered share* $\Delta_{\text{sel}}/(A_{20,8} - A_{5,32})$ relates the gain to the accuracy difference between random cohorts of twenty patients with eight patches and five patients with 32 patches in the breadth–depth grid. Second, each test patient receives per draw its coverage improvement, the distance to its nearest random training patient minus the distance to its nearest dispersed training patient, and Δ_{sel} is computed within tertiles of this improvement with the patient weights of the primary endpoint. Coverage predicts a gain that grows with the improvement. Third, all quantities are reported separately for the thirteen multi-site classes of Queisler (2026e) (cancer types with at least ten tissue source sites of at least four eligible patients each) and the seventeen remaining classes. Fourth, the between-patient similarity ω and the similarity-adjusted effective support N_{eff}^{ω} of both allocations are computed as in Queisler (2026b), and the relation fitted there to random cohorts is applied to both allocations. With the coverage distances of Table 2, its coverage term alone amounts to about 5.3 points. This value is a descriptive benchmark, not a forecast: the relation overpredicted the selection gain at twenty patients and has not been validated outside the range of random cohorts.

3.2 Uncertainty

Uncertainty follows Queisler (2026c). Each of 2,000 paired Bayesian bootstrap replicates assigns Dirichlet(1, ..., 1) weights to the distinct test patients (Rubin, 1981), shared across allocations, splits, draws, and classes. The 2.5th and 97.5th percentiles give exploratory 95% intervals. Embeddings and allocations are held fixed. The standard deviation of each allocation’s accuracy and of the contrast across the fifteen split–draw combinations is reported as a descriptive measure of sensitivity to the drawn patients.

4 Interpretation criteria

Conclusions rest on the test-patient interval for Δ_{sel} and on the practical threshold of one percentage point used in the preceding experiments (Table 3). Because the manipulation check guarantees a coverage change of about two thirds of the coverage gained from five to twenty random patients, an interval within $[-1, 1]$ counts as practical equivalence.

Reading	Claim	Selection gain Δ_{sel}
Selection mitigates shortage	Coverage-maximizing selection trains more accurate classifiers than random sampling of five patients	Interval above 1
No practical gain	A large coverage change leaves accuracy practically unchanged at five patients	Interval within $[-1, 1]$
Selection harms	Coverage-maximizing selection trains less accurate classifiers than random sampling	Interval below -1

Table 3: Each reading predicts a distinct range of the selection gain. Intervals are test-patient 95% intervals in percentage points. All other intervals, including those that cross a threshold, are inconclusive.

Selection mitigates shortage. If the gain exceeds the threshold, patient-mean coverage separates useful from less useful cohorts at a patient count other than twenty, and choosing patients by coverage recovers part of the breadth benefit when only five patients can be labelled. The recovered share and the tertile gradient would then show how much of the gap and which test patients the selection reaches. Before selection can be recommended, probability quality would need to be examined.

No practical gain. If the gain is practically zero, a coverage change of about two thirds of the coverage gained from five to twenty random patients does not change accuracy. Together with the concentration damage at twenty patients, this would indicate that poor coverage hurts but that better-than-random coverage does not help, so patient-mean coverage would be weak as a selection signal. The next candidate would be a distance between the patch distributions of patients.

Selection harms. If selection lowers accuracy, the selected patients cover validation patients in the embedding but not the variation that matters for the classifier, for example because greedy selection favours atypical patients. Coverage distance would then be a misleading target for patient selection, whatever its association with accuracy in random cohorts.

5 Results

Coverage-maximizing selection trains a more accurate classifier than random sampling of five patients. The selected cohort reaches 67.87 percent patient-macro balanced accuracy against 62.35 percent for the random cohort, a selection gain of $\Delta_{\text{sel}} = 5.52$ percentage points (95% interval: 4.83 to 6.23). The interval lies above the one-point threshold, so the outcome is the

reading *selection mitigates shortage* of Table 3. The manipulation check computed alongside the classifiers reproduces Table 2 exactly.

5.1 Selection gain at five patients

The dispersed cohort is more accurate than the random cohort in both class groups and less sensitive to which patients are drawn (Table 4). Across the fifteen split–draw combinations, its accuracy varies with a standard deviation of 1.14 points, against 1.40 points for the random cohort. The random cohort matches its counterpart on the breadth–depth grid, five random patients with 32 patches each at 62.25 percent, to 0.10 points (Queisler, 2026c). Five selected patients are about as accurate as ten random patients with sixteen patches each, which reach 67.28 percent at the same budget of 160 patches per class; their coverage distance of 0.324 is also slightly shorter than the 0.333 of ten random patients (Queisler, 2026b).

Allocation	All classes		Multi-site classes		Other classes	
	Accuracy [95% interval]	SD	Accuracy	SD	Accuracy	SD
Random	62.35 [61.15, 63.44]	1.40	60.19	2.10	64.01	1.75
Dispersed	67.87 [66.53, 69.13]	1.14	66.71	1.12	68.76	2.28

Table 4: At five patients with 32 patches each, the dispersed cohort is about five and a half points more accurate than the random cohort. Accuracy is patient-macro balanced accuracy in percent, averaged over three splits and five draws, with test-patient 95% intervals. Multi-site classes are the thirteen cancer types with at least ten tissue source sites of at least four eligible patients each (Queisler, 2026e); other classes are the remaining seventeen. SD is the population standard deviation across the fifteen split–draw combinations, in percentage points.

The selection gain is positive in all fifteen split–draw combinations, ranging from 2.20 to 7.94 points with a standard deviation of 1.62 points (Table 5). Averaged over draws, it is 4.80, 4.79, and 6.97 points in the three splits. Even the smallest gain of a single combination exceeds the threshold.

The gain recovers 0.58 of the accuracy difference between random cohorts of twenty patients with eight patches and five patients with 32 patches on the breadth–depth grid, 71.76 against 62.25 percent (95% interval: 0.51 to 0.65). Selection therefore closes more than half of the breadth advantage without adding a patient. At twenty patients, the same selection rule added 1.04 points over random sampling, with a coverage change of about a quarter of the coverage gained from five to twenty random patients (Queisler, 2026a). At five patients, the coverage change is about two thirds of that gain and the accuracy gain five times as large.

Classes	ρ_{sel}	Δ_{sel}	95% interval	SD	Positive	Recovered share
All	0.67	5.52	[4.83, 6.23]	1.62	15 of 15	0.58 [0.51, 0.65]
Multi-site	0.74	6.52	[5.85, 7.23]	2.15	15 of 15	0.70 [0.64, 0.77]
Other	0.62	4.75	[3.61, 5.86]	2.58	15 of 15	0.49 [0.38, 0.59]

Table 5: The selection gain lies above the one-point threshold in both class groups. ρ_{sel} is the selection ratio of Equation (4), computed from coverage distances averaged over the group’s classes, splits, and draws. Estimates and paired test-patient intervals are in percentage points. SD and the positive count describe the gain across the fifteen split–draw combinations. The recovered share divides the gain by the grid difference between twenty patients with eight patches and five patients with 32 patches within the same group, with its test-patient 95% interval.

5.2 Where the gain arises

Selection gains more in the multi-site classes, 6.52 points, than in the seventeen other classes, 4.75 points, and it recovers 0.70 against 0.49 of the grid difference (Table 5). The multi-site classes also have the larger coverage change, with a selection ratio of 0.74 against 0.62, and their random cohorts are the least accurate, at 60.19 percent. Tissue source sites do not account for the difference. In the multi-site classes, 7.7 percent of the four selected patients share the focal patient’s site, against 8.3 percent of the four other anchors; in the other classes, the shares are 25.1 and 22.6 percent.

Within classes, the gain falls on test patients whose coverage selection improves most (Table 6). In the tertile with the largest coverage improvement, recall rises from 54.25 percent under random training to 69.68 percent under dispersed training, a gain of 15.43 points. In the middle tertile, the gain is 5.33 points. In the tertile with the smallest improvement, dispersed training is 3.34 points less accurate than random training, so selection trades a small loss for patients whose coverage it improves least against a large gain for patients that the random cohort covers poorly. After selection, recall differs by less than seven points between the three tertiles, whereas under random training the tertile with the largest improvement trails the others by eleven to thirteen points. The gradient appears in both class groups.

Coverage-improvement tertile	All classes			Gain by class group	
	Random	Dispersed	Gain	Multi-site	Other
Low	67.07	63.74	−3.34	−2.36	−4.09
Middle	65.28	70.61	5.33	6.13	4.72
High	54.25	69.68	15.43	16.07	14.93

Table 6: The selection gain grows with the coverage that selection adds for a test patient. Tertiles split each class’s test patients per draw by their coverage improvement, the distance to the nearest random minus the distance to the nearest dispersed training patient. Recalls and gains are in percentage points, with the patient weights of the primary endpoint within a tertile and equal weight for each class, split, and draw. Estimates are descriptive and carry no intervals.

5.3 Relation fitted to random cohorts

The selected patients differ from one another more than random patients do (Table 7). Their mean between-patient similarity is $\omega = -0.094$, against -0.011 for the random cohort, so similarity-adjusted effective support rises from 12.13 to 17.88 while effective support stays at 10.06 for both cohorts. Averaged within each cell of the breadth–depth grid, random cohorts have a similarity between -0.013 and -0.010 (Queisler, 2026b).

The relation predicts a selection gain of 5.77 points (4.80 to 6.78), and its prediction error of 0.25 points has an interval of -0.87 to 1.43 . Coverage distance contributes 5.34 points of the prediction and similarity 0.43 points. The coverage part agrees with the descriptive benchmark of Section 3.1, which was fixed before any classifier was trained but not as a forecast.

At twenty patients, the same relation predicted 2.10 points against an observed 1.04 (Queisler, 2026b). The two settings differ in where the dispersed cohort’s coverage distance lies. Random cohorts of five to twenty patients span coverage distances from 0.288 to 0.398. Five selected patients, at 0.324, lie inside this span, whereas twenty selected patients, at 0.269, lie beyond it. The relation therefore interpolates at five patients and extrapolates at twenty.

Allocation	Observed A	Predicted A	N_{eff}^{ω}	ω	r
Random	62.35	62.12 [60.91, 63.23]	12.13	−0.011	0.386
Dispersed	67.87	67.89 [66.31, 69.41]	17.88	−0.094	0.324
Selection gain	5.52	5.77 [4.80, 6.78]			

Table 7: The relation fitted to random cohorts reproduces both allocations and the selection gain. The relation is model (c) without $\log G$ of Queisler (2026b), fitted to the 135 random cohorts of the breadth–depth grid and applied to the 30 cohorts of this experiment. A is patient-macro balanced accuracy in percent, with test-patient 95% intervals for predictions. N_{eff}^{ω} , ω , and r are means over three splits and five draws of the class-averaged similarity-adjusted effective support, between-patient similarity, and coverage distance.

6 Discussion and conclusion

When only five patients per cancer type can be labelled, which five matters. At the same patient count, patches per patient, budget, and effective support, five patients chosen by coverage-maximizing selection train a classifier 5.52 points more accurate than five randomly drawn patients. The gain appears in every split–draw combination and in both class groups, recovers 58 percent of the advantage of twenty over five random patients, and makes five selected patients about as accurate as ten random ones.

Three observations tie the gain to coverage. First, it concentrates on test patients whose distance to the nearest training patient shrinks most under selection, while patients already close to the random cohort lose a little. This is the mirror image of the concentration damage at twenty patients, which fell on test patients that a clustered cohort left uncovered (Queisler, 2026a). Second, a relation fitted only to random cohorts, in which coverage distance is nearly fixed by patient count, predicts the gain with an error interval that includes zero and attributes almost all of it to coverage. Third, the gain is large where random sampling leaves much to cover, at five patients, and small where it does not, at twenty. Together with the overprediction at twenty patients, this pattern suggests that accuracy responds to coverage distance roughly linearly over the range of random cohorts and flattens once coverage exceeds what random cohorts reach.

Coverage is not the only property that selection changes. Selected patients are mutually dissimilar, so they are less redundant with one another than random patients, and similarity-adjusted effective support rises by nearly half. The relation assigns only 0.43 points to this change, but its similarity coefficient is estimated from random cohorts, in which similarity barely varies and stays far above the value of selected cohorts. The tertile gradient favours coverage over redundancy, because a reduction of redundancy alone would not be expected to lower recall for the best-covered test patients, but it does not separate the two.

For patient shortage, the result shifts the question from how many patients to label to which patients to label. Selection needs only unlabelled embeddings of candidate patients, and at five patients it reaches accuracy that random sampling attains only with twice as many patients. Before it can be recommended, the criteria of Section 4 require a further check: probability quality, measured by negative log-likelihood and calibration error, must not deteriorate under selection.

7 Limitations

The two cohorts share the focal patient, so selection starts from a randomly drawn patient rather than from the best available one, and the gain is that of four selected patients. The manipulation is uneven across cancer types. The class-level ratio stays below 0.5 in four or five classes per

split, among them uveal melanoma, uterine carcinosarcoma, and adrenocortical carcinoma, in which the selected patients cover validation patients little better or even worse than the anchors; these classes contribute little contrast to the class average.

Selection draws from a pool of unlabelled candidate patients from the same TCGA-UT cohort as the test patients; in practice, candidates may come from a population that differs from the one the classifier later meets. Coverage and similarity are defined on patient means in one representation. Probability quality was not evaluated. All results are conditional on frozen Virchow2 features, multinomial logistic regression, and the three overlapping splits, which are not independent replications.

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